
Sarra BEN ABBES

Tunis, Tunisia

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Summary

IT engineer and aspiring researcher with a strong foundation in Information Technology and applied Machine Learning/Computer Vision. Experienced in aquaculture monitoring, where I developed PL counting and shrimp size/weight estimation systems and advanced the work to an IEEE conference submission. Skilled in addressing small-object detection in clustered and occluded environments, applying methods such as slicing strategies, post-processing, and structured evaluation. Adept at building end-to-end ML pipelines and aligning research with practical applications. Passionate about advancing AI to solve critical challenges in agriculture, aquaculture, and medicine.

Experience

Aqua Development Ltd

AI Engineer - Department of Research and Development • Tunis, Tunisia

05/2025 - Present

- Advancing post-larvae (tiny new born) shrimp counting project to a publishable level, refining detection methods and evaluation pipelines.
- Leading ongoing research on shrimp size and weight estimation, including dataset building and testing new approaches (multi-dimensional measurements, segmentation-based models, and integration with pond-level data).
- Designed and implemented slicing and post-processing strategies to improve small-object detection in clustered and occluded environments.
- Contributed to the preparation of a research paper submitted to IEEE (under review) and supporting ongoing publications.
- Collaborating closely with developers and farm experts to ensure solutions remain both scientifically rigorous and applicable in real-world aquaculture.

ROOLY S

IT Engineer and Multipotentialite • Tunis, Tunisia

03/2025 - 06/2025

- Took on cross-functional responsibilities across HR, marketing, business analysis, and business development, supporting the startup in multiple growth areas.
- Acted as an advisor to a machine learning intern, providing technical guidance and reviewing deliverables to align with project needs.
- Contributed to HR operations, including onboarding support and candidate assessments, ensuring smooth integration of new team members.
- Led the marketing department, analyzing the company's digital presence, creating a strategy and action plan, and executing targeted initiatives.
- Conducted a market analysis to identify industry needs, define main target sectors, and propose project ideas for prospection.

ROOLY S

Business Analyst Intern • Tunis, Tunisia

09/2024 - 11/2024

- Defined functional requirements, drafted user stories, and mapped workflows for a custom web application.
- Collaborated with developers and testers to ensure business needs were accurately translated into technical solutions.
- Participated in user training sessions and contributed to onboarding documentation.

Aqua Development Ltd.

Machine Learning Intern • Tunis,Tunisia

03/2024 - 07/2024

- Developed and optimized computer vision models (YOLOv8) for shrimp post-larvae counting and biometric sizing.
 - Designed and implemented complete inference pipelines, from model training to deployment on desktop (PyQt) and mobile (Flutter).
 - Applied size-to-weight regression models and integrated OpenCV-based preprocessing techniques.
 - Visualized results through annotated outputs and statistical graphs to support evaluation and analysis.
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Publications

Cross-Dataset Shrimp Larvae Counting via Tiled Inference and Domain Filtering

Sarra Ben Abbes, Divya Velayudhan, Naoufel Werghi

Submitted to: *IEEE International Conference on Artificial Intelligence, Data Analytics and Computational Intelligence Systems (ADACIS 2025)*.

Summary: This work introduces a lightweight vision-based detection framework for accurate counting of post-larval shrimp. By combining tiled (slicing-based) inference with domain-aware filtering, the method improves recall in high-density scenes while maintaining precision. Unlike prior approaches, it is evaluated under a cross-dataset protocol, demonstrating robustness across unseen datasets and real-world aquaculture conditions.

Manifestation-Aware Contrastive Learning for Cross-Domain Tomato Disease Classification with Limited Data

M. Shafay, D. Velayudhan, S. Ben Abbes, T. Hassan, I. Hussain, N. Werghi

Submitted to: *IEEE International Conference on Artificial Intelligence, Data Analytics and Computational Intelligence Systems (ADACIS 2025)*

Summary: This work introduces *Manifestation-Aware Contrastive Learning (MACL)*, a vision-language framework for plant disease classification under diverse field conditions. By combining multi-scale visual crops with diverse, symptom-focused textual descriptions, MACL captures both global and fine-grained disease manifestations. Tested across four unseen tomato disease datasets, MACL significantly outperforms state-of-the-art vision-language models (CLIP, LongCLIP, AgriCLIP), achieving stronger cross-domain generalization and interpretable disease localization.

Skills

Machine Learning, Deep Learning, Computer Vision, Artificial Intelligence, Team Management, Stakeholder Management, Leadership, Project Management

Extracurricular

Junior Enterprises Tunisia (JET) — 2 years

During my two years at JET, I progressed from **Chargée de Missions Qualité** to leadership roles such as **Quality Manager**, **Quality Auditor**, and **Vice-President**, gaining extensive experience in quality management, audits, organizational performance, and strategic development.

Key Roles & Achievements:

Quality Management & Audits

- Led and managed audit missions across Junior Enterprises, ensuring compliance with JET's internal quality standards.
- Improved the Quality Management System within JET and successfully maintained the **ISO9001 certification**.
- Conducted internal quality audits and contributed to the development of sustainable processes.

Leadership & Strategy (Vice-President)

- Piloted strategic processes in **Innovation, Sustainable Development, Skills Development, and Organizational/Financial Internal Control**.
- Oversaw performance initiatives and continuous improvement efforts across the network.
- Coordinated with teams to implement responsible business practices and drive innovation.

Skills Developed:

- Team Leadership & Management
 - Process Improvement & Continuous Improvement
 - Internal Controls & Audits
 - Corporate Social Responsibility
 - Project & Innovation Management
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Education

Ecole Nationale d’Ingénieurs de Carthage (ENICarthage)

Computer Science • Tunis,Tunisia 12/2024

Computer Science Engineering Degree – Diplome National d'Ingénieur

Esprit Prépa

Maths/ Physics • Tunis,Tunisia 04/2020

Engineering School Admission

Bourguiba Pioneer High School

Maths • Tunis,Tunisia 06/2018

Baccalaureate Degree

Languages

Arabic, English, French

Certificates

Digital Marketing from Hubspot, Google Analytics from Google